Journal Name

Crossmark

RECEIVED
dd Month yyyy

REVISED
dd Month yyyy

PAPER

Teacher-guided AI-assisted development of virtual physics laboratories: classroom case studies from secondary physics

Anurag Garg[1]

[1]Physics Department, Ministry of Education, United Arab Emirates
[1]Author to whom any correspondence should be addressed.

E-mail: anurag.garg@moe.sch.ae

Keywords: Physics education, Generative AI, Virtual laboratories, Single-slit diffraction, Transformer simulation, Human-in-the-loop pedagogy

Abstract
This classroom case study examines the teacher-guided AI-assisted development of virtual physics laboratory activities for secondary education. A generative AI system was used as a rapid prototyping partner to create and iteratively refine interactive simulations, digital measurement tools, and supporting laboratory worksheets for Grade 10 and Grade 12 physics. The first case study focused on the development of a single-slit diffraction laboratory emphasizing conceptual visualization, measurement calibration, and error analysis. A second case study involving transformer simulation development demonstrated how previously established design patterns and pedagogical expectations enabled substantially faster refinement using simpler prompts. Classroom implementation and exploratory student feedback are discussed in relation to engagement, usability, visualization, and perceived support for experimental understanding. The study highlights how generative AI may support rapid development of curriculum-linked virtual laboratories when the teacher remains central in directing pedagogy, verifying physics content, and adapting tools for learners.

1 Introduction

Interactive simulations and virtual laboratories have become increasingly important tools in physics education, particularly for topics involving abstract processes, dynamic systems, or experiments that are difficult to reproduce consistently in school laboratories. Simulations can support conceptual visualization, inquiry-based learning, and rapid exploration of relationships between physical variables. The educational value of interactive physics simulations has been demonstrated through projects such as PhET, which emphasized visualization, interactivity, and conceptual exploration in physics teaching [1, 2]. Broader reviews of digital games, simulations, and science education have also reported positive learning effects when simulations are carefully integrated with teacher support, lesson design, and inquiry activities [3, 4, 5, 6]. However, the development of custom educational simulations often requires programming experience, substantial preparation time, and repeated testing to ensure that the final tool remains pedagogically suitable for classroom use.

Within secondary physics curricula, topics such as single-slit diffraction require students to simultaneously interpret how wavelength (λ), slit width (a), and screen distance (D) influence the width of the central maximum. Although textbook diagrams provide static representations of diffraction patterns, students often have limited opportunities to dynamically manipulate these variables or conduct measurement-based exploration during regular classroom instruction. Consequently, converting a visualization into a usable classroom laboratory frequently requires additional features such as calibrated measurement tools, constrained physical parameter ranges, and worksheet integration.

Recent advances in generative artificial intelligence (AI) systems have introduced new possibilities for rapid prototyping of educational materials, including code generation, worksheet design, and interactive visualization tools. Current research in AI in education has explored both the opportunities and limitations associated with these technologies [8, 12]. Studies have also highlighted the evolving role of teachers in AI-supported educational design and implementation

[9]. At the same time, generative AI has raised new questions regarding instructional design, hybrid authorship, and collaborative human–AI workflows in educational settings [10]. Work on generative AI in software engineering education further suggests that code-generation tools can support practical prototyping when combined with appropriate verification and instructional oversight [11]. Despite growing interest in AI-assisted educational development, comparatively fewer studies have examined how physics teachers may iteratively collaborate with generative AI systems to create classroom-oriented learning tools while retaining pedagogical control over the design process.

This paper presents two classroom case studies involving teacher-guided AI-assisted development of virtual physics laboratory activities for secondary education. In the first case, a Grade 10 single-slit diffraction simulation was iteratively developed to support visualization, measurement, and error analysis during a diffraction laboratory activity. The development process involved repeated refinement of fringe visibility, parameter ranges, measurement calibration, and digital worksheet integration to align the simulation with classroom requirements. In the second case, a Grade 12 transformer simulation was developed using a substantially shorter prompting process after design patterns and pedagogical expectations had already been established through the earlier interaction. Together, these cases provide insight into how generative AI may function as a rapid prototyping partner during simulation development while highlighting the continuing importance of teacher guidance, physics verification, and classroom adaptation.

Rather than treating AI as an autonomous educational designer, this study examines a collaborative human-in-the-loop workflow in which the teacher repeatedly evaluated outputs, corrected unsuitable features, refined the classroom usability of the simulations, and integrated the resulting tools into curriculum-linked learning activities. This framing is important because AI-generated code may still require expert checking, debugging, and validation before classroom use, particularly in physics contexts where non-physical assumptions or incorrect parameter choices can undermine learning [13, 14]. The paper focuses on the development workflow, classroom implementation, and exploratory student feedback associated with these AI-assisted virtual laboratories.

2 Method: The AI-Assisted Development Process

The development workflow used across the classroom case studies followed an iterative, teacher-guided, human-in-the-loop (HITL) engineering process consisting of four recurring stages: initial conceptual prompting, AI generation, teacher-led pedagogical verification, and local iterative refinement. Rather than treating the generative AI system as an autonomous curricular designer, it was used as a rapid prototyping partner [**?**] that produced editable simulations, worksheets, and interface components in response to natural language instructions. This workflow aligns with emerging frameworks in educational technology that emphasize the necessity of keeping the educator central to auditing and anchoring AI-generated content within localized school contexts [18, 15].

2.1 Initial prompting and context setting

In the first stage, the teacher provided open-ended prompts describing the target physics concept, intended classroom use, and desired student interaction. These prompts generally specified the educational objective rather than detailed programming instructions, utilizing natural language as an abstraction layer over source code [19]. In the Grade 10 diffraction case, the initial prompt asked for a simulation showing how wavelength (λ), slit width (a), and screen distance (D) affect the diffraction pattern and the width of the central maximum (W). Later prompts requested classroom-specific features such as improved fringe visibility, a digital measurement ruler, and worksheet integration. In the Grade 12 cases, the prompts were shorter and more operational, asking for simulations in which transformer voltage changed with primary and secondary turns, or in which LCR circuit quantities could be varied and observed.

2.2 AI generation and script prototyping

The second stage involved AI generation of standalone HTML, CSS, JavaScript, and fillable LaTeX resources. The generated outputs included interactive simulations, digital rulers, animated visualization elements, and laboratory worksheets. The simulation scripts implemented standard school-level physics relationships at runtime to drive real-time graphical rendering [**?**]. For the diffraction simulation, the generated code used the single-slit diffraction intensity relation,

$$\beta = \frac{\pi a \sin\theta}{\lambda}, \qquad I(\theta) = I_0 \left(\frac{\sin\beta}{\beta}\right)^2. \tag{1}$$

For the transformer simulation, the output voltage was linked to the turns ratio,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}, \tag{2}$$

while the LCR activity used standard expressions for inductive reactance, capacitive reactance, and impedance,

$$X_L = \omega L, \qquad X_C = \frac{1}{\omega C}, \qquad Z = \sqrt{R^2 + (X_L - X_C)^2}. \tag{3}$$

These formulae were not treated as novel contributions; rather, they provided the physics backbone for classroom visualizations and real-time numerical displays.

2.3 Teacher evaluation and scientific auditing

The third stage consisted of teacher evaluation, scientific checking, and classroom usability review. Left unchecked, generative AI tools frequently introduce subtle scientific errors, non-physical boundary assumptions, or operational constraints that neglect target student math levels [20]. Therefore, generated outputs were tested locally before classroom use and were checked for physics accuracy, visual clarity, curriculum alignment, and practical deployability. In the early diffraction development cycle, a local execution issue occurred when saving HTML code through macOS TextEdit, because rich-text formatting could prevent the browser from interpreting the file as executable HTML. This led to revised deployment instructions emphasizing plain-text saving and browser-based execution.

The teacher also audited parameter boundaries and visual scaling. In the diffraction case, the simulation required adjustment of slit-width and screen-distance ranges to avoid patterns that were either too compressed to measure or so wide that the first minima moved outside the visible screen. Additional refinements included brightness enhancement for secondary fringes, calibrated ruler placement, and a zoom adjustment so that students could measure the central maximum meaningfully. In the LCR activity, input intervals for inductance and capacitance were adjusted so that resonance and impedance changes remained observable within the intended frequency range.

2.4 Iterative refinement and warm-start development

The final stage involved iterative refinement through continued teacher–AI interaction. The Grade 10 diffraction simulation functioned as a cold-start case: it required several interaction cycles to move from a technically functional visualization to a classroom-ready virtual laboratory with measurement and worksheet components. Through this process, recurring design preferences were established, including range-slider controls, clear numerical readouts, visual emphasis on observable effects, and accompanying fillable worksheet structures—features recognized for reducing cognitive load during virtual inquiries [2].

When the same extended AI interaction context was later used to create Grade 12 transformer and LCR simulations, substantially shorter prompts produced usable initial outputs. This is not interpreted as permanent model learning. Rather, it suggests that maintaining conversational continuity allowed previously established design patterns and pedagogical expectations to influence later outputs, a phenomenon known in prompt engineering as utilizing in-context learning through conversation state preservation [**?**]. For example, the transformer activity quickly adopted a familiar structure with voltage readouts, turn-number sliders, canvas-based visual representation, and worksheet tasks involving power and efficiency,

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%. \tag{4}$$

Both case studies therefore operated within a collaborative human-in-the-loop framework in which the teacher retained responsibility for pedagogical direction, scientific verification, classroom adaptation, and final deployment decisions [18]. The AI system accelerated prototyping and code generation, while the teacher continuously filtered, corrected, and refined the outputs before classroom use.

Acknowledgments

Sample text inserted for demonstration.

Funding

Sample text inserted for demonstration.

Table 1. Summary of the teacher–AI co-development process across the classroom cases.

Case	Physics focus	Development pattern	Main pedagogical refinement
Grade 10	Single-slit diffraction	Cold-start iterative development	Fringe visibility, calibrated ruler, measurement and error-analysis worksheet
Grade 12	Transformer induction	Warm-start development	Voltage-ratio visualization, turns sliders, power and efficiency analysis
Grade 12	Series LCR circuit	Warm-start development	Real-time reactance, impedance, and resonance displays

Author contributions
Sample text inserted for demonstration.

Data availability
Sample text inserted for demonstration.

Supplementary data
Sample text inserted for demonstration.

References
References

[1] Finkelstein N, Adams W, Keller C, Perkins K and Wieman C 2006 High-Tech Tools for Teaching Physics: The Physics Education Technology Project *Journal of Online Learning and Teaching* **2**(3)

[2] Wieman C E, Adams W K, Loeblein P and Perkins K K 2010 Teaching Physics Using PhET Simulations *The Physics Teacher* **48**(4) 225–227 doi:10.1119/1.3361987

[3] Clark D B, Tanner-Smith E E and Killingsworth S S 2015 Digital Games, Simulations, and Learning: A Systematic Review and Meta-Analysis *Review of Educational Research* **86**(1) 79–122

[4] Rutten N, van Joolingen W R and van der Veen J T 2012 The learning effects of computer simulations in science education *Computers & Education* **58**(1) 136–153 doi:10.1016/j.compedu.2011.07.017

[5] Smetana L K and Bell R L 2012 Computer Simulations to Support Science Instruction and Learning: A critical review of the literature *International Journal of Science Education* **34**(9) 1337–1370 doi:10.1080/09500693.2011.605182

[6] Banda H 2021 Effect of integrating physics education technology simulations on students' conceptual understanding in physics: A review of literature doi:10.1103/PHYSREVPHYSEDUCRES.17.023108

[7] Sellberg C and Lindwall O 2026 Simulation-based training in professional education: learning, participation, and instructional design *Instructional Science* **54** 9 doi:10.1007/s11251-025-09763-2

[8] Holmes W and Tuomi I 2022 State of the Art and Practice in AI in Education *European Journal of Education* **57**(4) 542–570 doi:10.1111/ejed.12533

[9] Celik I, Dindar M, Muukkonen H and Jarvela S 2022 The Promises and Challenges of Artificial Intelligence for Teachers: A Systematic Review of Research *TechTrends* **66**(4) 616–630 doi:10.1007/s11528-022-00715-y

[10] Hodges C B and Kirschner P A 2024 Innovation of Instructional Design and Assessment in the Age of Generative Artificial Intelligence *TechTrends* **68** 195–199 doi:10.1007/s11528-023-00926-x

[11] Li Y, Keung J and Ma X 2024 Integrating Generative AI in Software Engineering Education: Practical Strategies *2024 International Symposium on Educational Technology (ISET)* 49–53 doi:10.1109/ISET61814.2024.00019

[12] Baidoo-Anu D and Ansah L O 2025 Education in the Era of Generative Artificial Intelligence (AI): Understanding the Potential Benefits of ChatGPT in Promoting Teaching and Learning *SSRN* Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4337484

[13] Huang T, Ren Z, Huang Y et al. 2026 Hallucination detection in LLM code generation: A sampling-based consensus verification approach *Automated Software Engineering* **33** 70 doi:10.1007/s10515-026-00605-0

[14] Lazaros K, Vrahatis A G and Kotsiantis S 2026 Human-in-the-Loop Artificial Intelligence: A Systematic Review of Concepts, Methods, and Applications *Entropy* **28**(4) 377 doi:10.3390/e28040377

[15] United States Office of Educational Technology 2023 *Artificial Intelligence and the Future of Teaching and Learning: Insights and Recommendations* (Washington, DC: U.S. Department of Education) Available at: https://digital.library.unt.edu/ark:/67531/metadc2114121/

[16] Xing Z, Liu Y, Cheng Z, Huang Q, Zhao D, Sun D and Liu C 2025 When Prompt Engineering Meets Software Engineering: CNL-P as Natural and Robust APIs for Human-AI Interaction *arXiv:2508.06942*

[17] Brown T B et al. 2020 Language Models are Few-Shot Learners *Advances in Neural Information Processing Systems* **33** 1877–1901

[18] Lazaros K, Vrahatis AG, Kotsiantis S. Human-in-the-Loop Artificial Intelligence: A Systematic Review of Concepts, Methods, and Applications. Entropy. 2026; 28(4):377. https://doi.org/10.3390/e28040377

[19] Suh S, Jiang E and Chang M 2023 Prompting as an Alternative Layer of Software Abstraction: Principles for Human-AI Co-Design *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST '23)* 304–318 doi:10.1145/3586183.3606732

[20] Liu F, Liu Y, Shi L, Yang Z, Zhang L, Lian X, Li Z and Ma Y 2026 Beyond Functional Correctness: Exploring Hallucinations in LLM-Generated Code *arXiv preprint arXiv:2404.00971*